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Special Issue Call for Papers - Digital Transformation of Higher Education: Pedagogical Innovation Toward Education 4.0?

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Abstract:

This special issue welcomes submissions that are concerned with the digital transformation of higher education, with an emphasis on pedagogical innovation in a dynamic socio-technical higher education landscape.

Keywords: Digital Transformation, Education 4.0, Pedagogy, Pedagogical Innovation, Socio-Technical, Information Systems.

Call for Papers

Higher education (HE) has been undergoing a “rapid digital transformation” since the COVID-19 pandemic (Nurhas et al., 2022, p. 2924). This transformation is represented by a shift to online learning and a greater reliance on digital and educational technologies (edtech), which have reshaped and materially altered learning and teaching environments (Zha & He, 2021). Parallel developments in learning analytics (Nguyen et al., 2021) and the rise of the so-called Education 4.0 technologies are expected to further influence existing conceptions of learning and teaching in HE (Mukul & Büyüközkan, 2023).

Education 4.0 denotes a period of education that leverages technologies associated with the fourth industrial revolution (Industry 4.0), such as artificial intelligence (AI), the Internet of Things (IoT) and Augmented / Virtual / Extended Reality (AR, VR and XR), to support personalised, experience-based education (Mukul & Büyüközkan, 2023). This period can be distinguished from others based on philosophical underpinnings such as: cybergogy; learning outcomes that incorporate hard and soft skills development; information largely derived from online sources; approaches that are student-centred; and technologies that allow for increased “connectivity, digitalization and virtualization” (Miranda et al., 2021, p. 2). Education 4.0 additionally promotes models of active learning through innovation in digital pedagogy, which requires deliberate design and advancement across four components, namely, competencies, methods, infrastructure, and technologies (Miranda et al., 2021).

Socio-Technical Change Toward Education 4.0 & the Need for Pedagogical Innovation

The transformation toward Education 4.0 calls for a nuanced understanding of technological developments and socio-technical change (Bearman & Ajjawi, 2023). Drawing on the transition to online learning during the COVID-19 pandemic as a case in point, socio-technical change within the HE ecosystem called into question entrenched methods and assumptions about learning and teaching, resulting in the need to clarify terminology regarding modes of education and pedagogy, reassessing student-educator interactions and instituting myriad adjustments to processes and practices (Bryson and Andres, 2020).

Since the pandemic and considering Education 4.0 technologies such as AI more specifically, it is expected that HE institutions will further develop and refine institutional processes and adjust digital innovation management activities (Truong & Papagiannidis, 2022). A balanced approach to the digital transformation of HE would compel institutions to capitalise on the potential benefits of AI while mitigating and addressing the anticipated risks and challenges (Bansal et al., 2024; Dwivedi et al. 2023). Attaining the vision of Education 4.0 through AI and related Industry 4.0 technologies prompts many questions. These relate to assessment design, performance measurements, the practical integration of Generative AI tools, and effective communication with students and other stakeholders in the HE ecosystem (Peres et al., 2023). Other matters that warrant attention include the possible alleviation of administrative burdens (Barros et al., 2023), the transformation of student support mechanisms (Chen et al., 2023), and the reform of academic integrity and assessment-related functions (TEQSA, 2024). In a study examining Generative AI in information systems (IS) education, the possible impact scenarios are reported as oscillating between minimal or no impact on IS faculty and practitioners resulting from AI use/integration, through to extreme impact scenarios, where individuals are rendered redundant (Van Slyke et al., 2023). These considerations are not isolated to the IS discipline; rather, they permeate disciplinary boundaries such that they challenge existing pedagogical approaches, which may require innovation.

Pedagogical innovation refers to the deliberate introduction of novel teaching methods and approaches to improve or enhance the student learning experience within a given educational context (Walder, 2024). In recent years, innovation in pedagogy has denoted the augmentation of existing HE processes and structural elements in a purposeful manner by fostering a “digital learning space” defined by pedagogical, technical, and organisational dimensions (Bygstad et al., 2022, p. 3). Within this setting, a socio-technical systems framing and perspective (see Abbas and Michael, 2022 for a review of socio-technical systems theory) can support the introduction of pedagogies that prepare students for complex environments characterised by both risk and ambiguity (Bearman & Ajjawi, 2023). The socio-technical perspective also has the potential to improve quality culture within HE institutions, through a recognition of the design considerations relevant to social elements such as competencies and collaboration, and technical considerations such as the systems in place, both of which need to be supported by values-based leadership (Legemaateet al., 2022). These socio-technical considerations are beneficial in view of encapsulating the multifaceted nature of Education 4.0, specifically the components that span social and technical dimensions (see Miranda et al., 2021 for an overview of the distinct components).

Information Systems as a Leader of the Digital Transformation of Education

The IS discipline is in a prime position to lead, operationalise, and substantially advance the Education 4.0 agenda, while potentially playing a pivotal role in shaping and underpinning the digital transformation of HE. This is based predominantly on the driving role of IS as an applied socio-technical discipline, where an integrative, systems-based approach is prominent (Van Slyke et al., 2021) and the interdisciplinary underpinnings of the discipline are apparent when compared with other computing disciplines (Hol et al., 2024). The IS discipline furthermore allows for the interconnected elements of “contextualisation, communication, and practice” to be addressed in explicit use cases, such as AI in the frame of digital agency and digital practice (Ågerfalk, 2020, p. 1). The significance of IS in the digital transformation of HE could also be attributed to the demonstrated role and relevance of IS in HE during and since the COVID-19 pandemic (Van Slyke et al., 2021). As such, IS faculty and practitioners could meaningfully contribute to pedagogical innovation, and the learning and teaching space more broadly, both within and external to the discipline.

Contributions could include drawing on existing approaches to investigating and operationalising responsibility and ethics in IS education (see El Idrissi et al. 2023 for a series of design guidelines for responsible innovation training and skills development of IS students and Grøder et al. 2022 for an example of responsible AI in IS education), in addition to promoting teaching that connects with practice and workforce needs (Mitchell, 2022) within IS and across other disciplines. Operationalising responsibility and ethics in IS curricula furthermore justifies new approaches to flexible education focusing on educators’ digital competencies (Núñez-Canal et al., 2022).

From a learner’s perspective, and concerning pedagogical agents, approaches that integrate societal, ethical, and political perspectives should also be designed in alignment with the values of learners (Richards and Dignum, 2019), whilst being sensitive to the degree of digital readiness with respect to specific learning activities (Zha & He, 2021). In keeping with the core objectives of Education 4.0 regarding skills development, students must also develop cognitive, interpersonal, and socio-technical skills – the skills that reflect the key drivers of the digital transformation of HE against an Industry 4.0 backdrop (Oliveira & De Souza, 2022) and that allow for the honing of soft skills such as information literacy (Wiley & Avery, 2023). The IS discipline is uniquely placed to guide socio-technical processes that contribute to the digital transformation of HE, to ensure that responsible, ethical, values- and skills-oriented approaches are employed within an Education 4.0 setting.

Special Issue Themes

This special issue seeks to explore the digital transformation of HE, focusing on the impact and implications of digital technologies as they relate to pedagogical innovation in an evolving, complex socio-technical environment. We invite submissions that address the following themes:

1. Digital transformation and digital disruption in the context of higher education
2. The role and impacts of digital technologies, such as artificial intelligence (AI), the Internet of Things (IoT) and Metaverse/Extended Reality (XR), on higher education pedagogy and pedagogical innovation
3. The research-teaching nexus in IS, reflecting the link between research and teaching practice concerning digital technologies
4. Higher education curricula that demonstrate pedagogical innovation through the integration of the socio-technical, ethical, and regulatory implications of digital technologies in learning and teaching environments
5. Education 4.0 in higher education, both in general and with respect to specific programs that integrate with Industry 4.0 technologies
6. Responsible innovation (ecosystems) in higher education, particularly in IS
7. Pedagogical innovation in IS curricula at both undergraduate and postgraduate levels in response to digital technologies
8. Research- and practice-led teaching in IS relevant to Education 4.0 and the digital transformation of higher education

9. The role of the IS discipline in informing/guiding/contributing to learning and teaching practice and the digital transformation of higher education
10. Pedagogical innovation through the integration of scholarly, industry, and government perspectives
11. Digital readiness and the pedagogical link in higher education
12. Digital literacy (e.g. AI literacy) in higher education
13. Critical perspectives on the theme of the Special Issue, including a) the digital transformation of higher education, b) pedagogical innovation, and c) Education 4.0.
14. Future-oriented studies that explore outlooks beyond Education 4.0

The guest editors may also approve other themes related to the digital transformation of higher education, Education 4.0, socio-technical HE ecosystems, and pedagogical innovation, including (geographically) diverse higher education ecosystems perspectives.

Submission Requirements

Manuscripts submitted to this special issue should follow the CAIS style guide, which can be accessed via the following webpage, which contains the CAIS Author Template: https://aisel.aisnet.org/cais/format.html. Formatted manuscripts should be submitted to the CAIS ScholarOne webpage, available at: https://mc.manuscriptcentral.com/cais. When uploading your manuscript to the CAIS ScholarOne site, please indicate that you are submitting to this special issue during the submission process.

Special Issue Dates

- Submission deadline: 30 June 2025
- First round notification: 15 September 2025
- Invited revisions deadline: 15 November 2025
- Second/final editorial decision: 15 January 2026
- Projected Publication: April 2026

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